Relativistic Statistical Mechanics

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1 Transformation Laws Table

The table 1 shows the results of the literature [1–5] and the results of this work for the transformation laws of the thermodynamic variables. Our results agree with [5] and partially agree with [3, 4].

	Einstein, Planck. [1, 2]	**Ott, Arzelièz.** [3, 4]	**Sutcliffe** [5]	**Guarín-Carvajal-Páez-Gutiérrez**
Pressure	$P = P_0$		$P_m = P_0$, $P_{th} = P_0\gamma^2$	$P = P_0\gamma^2$
Volume	$V = V_0/\gamma$			
Entropy	$S = S_0$			
Temperature	$T = T_0/\gamma$	$T = T_0\gamma$		
Heat	$\delta Q = \delta Q_0/\gamma$	$\delta Q = \delta Q_0\gamma$		
Internal energy	$E = E_0\gamma + P_0V_0\gamma v^2/c^2$	$E = E_0\gamma$		
Helmholtz free energy	$F = F_0/\gamma$	$F = F_0\gamma$		

Table 1: Transformation of the thermodynamic quantities for different authors. This table was adapted from [6]. A zero subindex indicates variables measured for a rest observer with the container of the gas, by convention $\gamma = 1/\sqrt{1 - v^2/c^2}$ is the Lorentz factor, P_m is the *mechanical pressure*, and P_{th} the *thermodynamic* one (as discussed by [5]).

2 The invariance of entropy

Theorem 1. The entropy $S = k_B \ln \Omega$ for a moving inertial observer K with respect to the container of an ideal gas (N equal particles of same mass, elastic collisions between particles and walls, and non-interacting particles) is equal to the entropy $S_0 = k_B \ln \Omega_0$ for a rest observer K_0 with the container.

Here, $\Omega = \Omega(E, V, N)$ is the number of states for K depending on the internal energy, the volume, and the number of particles respectively. Analogously, $\Omega_0 = \Omega_0(E_0, V_0, N_0)$ is the number

of states for K_0 depending on its corresponding internal energy, volume and number of particles respectively. Besides, the number of particles for each frame must be the same, this is $N = N_0$.

Proof:
By definition

$$\Omega = \frac{1}{N!}\frac{1}{h^{3N}} \underbrace{\int_{\mathscr{V}} d^3r_1 \cdots d^3r_N}_{\Omega_v} \underbrace{\int_{\mathcal{R}} d^3p_1 \cdots d^3p_N}_{\Omega_p} , \tag{1}$$

where h is the planck constant, $\mathbf{r}_i$ and $\mathbf{p}_i$ are the 3−position and 3−momentum of the ith particle for K, $\mathscr{V}$ is the volume space, and $\mathcal{R}$ is the momentum space defined as

$$\mathcal{R} = \left\{ (\mathbf{g}_1, \ldots, \mathbf{g}_N) \in \mathbb{R}^{3N} \mid \sum_{i=1}^{N} c\sqrt{m^2c^2 + |\mathbf{g}_i|^2} = E_0 \right\} , \tag{2}$$

where $\mathbf{g}_i$ is the 3−momentum of the ith particle according to K_0. Now, by the non-interacting condition, the position $\mathbf{r}_i$ of each particle is independent one of the other. And therefore we can separate Ω_v as a product of integrals

$$\Omega_v = \left(\int_V d^3r_1 \right) \cdots \left(\int_V d^3r_N \right) = \left(\int_V d^3r_1 \right)^N = V^N . \tag{3}$$

Lemma 1. The volume V_0 for the observer K_0 changes to $V_0/\gamma = V$ by a Lorentz transformation corresponding to a Lorentz Boost in one direction. γ is the Lorentz factor.

By the previous lemma

$$\Omega_v = \frac{{V_0}^N}{\gamma^N} = \frac{\Omega_{v_0}}{\gamma^N} . \tag{4}$$

By the other side, chosen an appropriate coordinate axis we can say that the Lorentz Boost occur in the x−axis. According to [8] the transformation laws for momentum in this case are

$$p_{x_i} = \gamma \left(g_{x_i} + \frac{v}{c^2} E_{0_i} \right) , \tag{5}$$

$$p_{y_i} = g_{y_i} , \tag{6}$$

$$p_{z_i} = g_{z_i} , \tag{7}$$

where $\mathbf{p}_i$ is the momentum of the ith particle for observer K, and $\mathbf{g}_i$ is the momentum of the same particle but for K_0. Besides, v is the relative speed between the observers, $\gamma = 1/\sqrt{1 - v^2/c^2}$, c is the speed of light, and $E_{0_i} = c\sqrt{m^2c^2 + {g_i}^2} = mc^2\sqrt{1 + \frac{{g_i}^2}{m^2c^2}}$ is the relativistic energy of the particle for K_0.

We want to implement a change of variables from p_i to g_i in the momentum integrals. The Jacobian of the transformation is

$$J(\mathbf{g}) = \frac{\partial p_{x^i}}{\partial g_{x^i}} = \begin{vmatrix} \frac{\partial p_x}{\partial g_x} & \frac{\partial p_x}{\partial g_y} & \frac{\partial p_x}{\partial g_z} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{\partial p_x}{\partial g_x} = \frac{\partial}{\partial g_x} \left[\gamma \left(g_x + \frac{v}{c^2} mc^2 \sqrt{1 + \frac{g^2}{m^2c^2}} \right) \right] . \tag{8}$$

Knowing that $g^2 = {g_x}^2 + {g_y}^2 + {g_z}^2$,

$$J(\mathbf{g}) = \gamma \left(1 + \frac{v}{c^2} mc^2 \frac{g_x/(m^2c^2)}{\sqrt{1 + \frac{g^2}{m^2c^2}}} \right) = \gamma \left(1 + \frac{v}{c} \frac{g_x/(mc)}{\sqrt{1 + \frac{g^2}{m^2c^2}}} \right) . \tag{9}$$

Then by implement this coordinate transformation for all the variables of the N particles one gets

$$\Omega_p = \int_{\mathcal{R}} d^3p_1 \cdots d^3p_N = \int_{\mathcal{R}} \gamma^N \prod_{i=1}^{N} \left(1 + \frac{v}{c} \frac{g_{x_i}/(mc)}{\sqrt{1 + \frac{{g_i}^2}{m^2c^2}}} \right) d^3g_1 \cdots d^3g_N \, . \tag{10}$$

On the other side, one can check that

$$\begin{aligned} \prod_{i=1}^{N}(1 + M_i) &= 1 + M_1 + M_2 + \cdots + M_N + \\ &\quad + M_1M_2 + M_1M_3 + \cdots + M_1M_N + \cdots M_{N-1}M_n + \\ &\quad + M_1M_2M_3 + \cdots M_1M_{N-1}M_N + \cdots M_{N-2}M_{N-1}M_N + \cdots , \\ &= 1 + \sum_{k=1}^{N} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le N} M_{i_1} \cdots M_{i_k} \, , \end{aligned} \tag{11}$$

where $\sum_{1 \le i_1 < i_2 < \cdots < i_k \le N}$ stands for the sum over all integers $i_1, \ldots, i_k$ that satisfies $1 \le i_1 < i_2 < \cdots < i_k \le N$. Therefore

$$\Omega_p = \int_{\mathcal{R}} \gamma^N \left(1 + \sum_{k=1}^{N} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le N} \frac{v^k}{c^k} \frac{g_{x_{i_1}}/(mc)}{\sqrt{1 + \frac{{g_{i_j}}^2}{m^2c^2}}} \cdots \frac{g_{x_{i_k}}/(mc)}{\sqrt{1 + \frac{{g_{i_k}}^2}{m^2c^2}}} \right) d^3g_1 \cdots d^3g_N \, . \tag{12}$$

Now, as integration is a linear operation,

$$\Omega_p = \gamma^N \int_{\mathcal{R}} d^3g_1 \cdots d^3g_N + \gamma^N \sum_{k=1}^{N} \sum_{1 \le i_1 < i_2 < \cdots < i_k \le N} \frac{v^k}{c^k} \int_{\mathcal{R}} \frac{g_{x_{i_1}}/(mc)}{\sqrt{1 + \frac{{g_{i_j}}^2}{m^2c^2}}} \cdots \frac{g_{x_{i_k}}/(mc)}{\sqrt{1 + \frac{{g_{i_k}}^2}{m^2c^2}}} d^3g_1 \cdots d^3g_N \, . \tag{13}$$

While integrating the second term we will find that if a point $(\mathbf{g}_1, \ldots, \mathbf{g}_k)$ is in $\mathcal{R}$, then $(-\mathbf{g}_1, \mathbf{g}_2, \ldots, \mathbf{g}_k)$ also will be in $\mathcal{R}$. This can be seen by looking in the definition of the momentum space 2. Therefore, $\mathcal{R}$ is composed by pairs of points $(\mathbf{g}_1, \ldots, \mathbf{g}_N)$ and $(-\mathbf{g}_1, \mathbf{g}_2, \ldots, \mathbf{g}_N)$ such that its respective contribution to the integral cancel out each other, this is

$$\int_{\mathcal{R}} \frac{g_{x_{i_1}}/(mc)}{\sqrt{1 + \frac{{g_{i_j}}^2}{m^2c^2}}} \cdots \frac{g_{x_{i_k}}/(mc)}{\sqrt{1 + \frac{{g_{i_k}}^2}{m^2c^2}}} d^3g_1 \cdots d^3g_N = 0 \, . \tag{14}$$

Then only the first term in 13 remains

$$\Omega_p = \gamma^N \int_{\mathcal{R}} d^3g_1 \cdots d^3g_N \, . \tag{15}$$

Regardless the factor γ^N, this integral is exactly the definition of Ω_p but for K_0. Hence $\Omega_p = \gamma^N \Omega_{p_0}$, and so $\Omega = \Omega_0$. This ends the proof.

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